50% which is the exact distribution of the reference dataset. However, the amount of drift increases as the percentage of positive class gets more different than the reference distribution. All encoders performed comparably but the sentence BERT and larger transformer models tend to detect drift faster due to producing higher quality text representations.

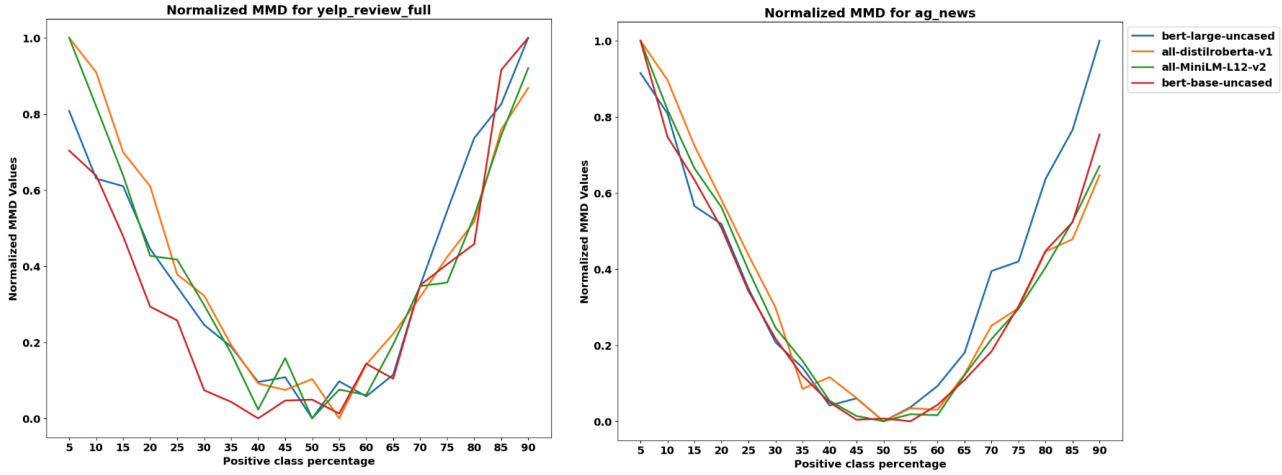

Figure 3: Left and right plots shows MMD values as we change the percentage of positive class in the target data resulting in data drift.

4 Discussion and Conclusion

Our proposed unsupervised drift detection approach for unstructured data, specifically text data, presents a novel and effective method for identifying potential drift in machine learning models. By utilizing maximum mean discrepancy (MMD) kernel-based statistical test and bootstrap, we can accurately identify the subset of data likely to be the root cause of the drift and suggest its adding back to the training data as a mitigation strategy. Our approach is particularly useful in identifying model performance regression and provide a fast mitigation solution for recovery.

In this paper, we demonstrated the efficacy of our proposed method for two use cases, but the approach can be applied to any other domain or dataset. Overall, our proposed method provides a robust framework for detecting and mitigating performance regression in machine learning models caused by drift.

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